Supplementary Material

Radiomics-Based Machine Learning and Deep Learning for Clinically Significant Prostate Cancer Detection on Biparametric MRI

S1 Sensitivity analysis: optimistic upper bound for the deep models under test-fold epoch selection

A natural concern when classical and deep models are compared is whether the deep architectures were disadvantaged by their training protocol. In the main analysis, the number of training epochs for each deep model was selected on an internal 80:20 split of the outer-training partition, after which the model was refit on the complete outer-training partition; the held-out outer fold was never used for any fitting or model-selection decision, exactly mirroring the leakage-safe treatment of the classical ensembles. To bound how much the deep models could gain from a more permissive—but optimistically biased—protocol, we repeated the deep-model experiments while selecting the training epoch by monitoring the held-out outer fold itself (up to 300 epochs). This procedure deliberately leaks information from the evaluation fold into model selection and therefore yields an *upper bound* on deep-model performance rather than a valid estimate of generalization; the classical models were left unchanged and serve as the leakage-safe reference. All metrics are pooled out-of-fold values with patient-level cluster-bootstrap 95% confidence intervals, reported at the fixed 0.5 probability threshold.

Even under this favorable protocol, the deep models did not overturn the main conclusions (Table S1). In the radiomics-only setting—the primary comparison—the leakage-safe Random Forest (AUROC 0.814) remained at least as discriminative as every test-fold-selected deep model (best deep AUROC 0.794), so the classical ensemble retained its advantage despite the deep models being granted a peek at the evaluation fold. Only in the combined conditions did the optimistically tuned deep models edge ahead of the Random Forest, and then only marginally and with broadly overlapping confidence intervals: by +0.009 AUROC for the concatenation Transformer (0.837 vs. 0.828) and by +0.015 AUROC for the dual-branch Transformer (0.843 vs. 0.828). These small, leakage-inflated margins are consistent with the main finding that, on this engineered radiomic feature space, deep tabular models do not confer a robust discrimination advantage over a calibrated tree ensemble, and they reinforce rather than weaken that conclusion: the deep models fail to clearly surpass the classical baseline even when the evaluation protocol is tilted in their favor.

Table S1: Optimistic upper bound for the deep models when the training epoch is selected on the held-out outer fold (a deliberately leakage-prone protocol), compared with the leakage-safe Random Forest reference carried over unchanged from the main analysis. Threshold-free metrics on pooled out-of-fold predictions with patient-level cluster-bootstrap 95% confidence intervals; lower Brier is better. Deep-model rows are optimistic upper bounds and are *not* valid estimates of generalization.

Condition	Model (protocol)	AUROC	AUPRC	Brier
Radiomics-only	Random Forest (leakage-safe, reference)	0.814 (0.790–0.837)	0.605 (0.561–0.655)	0.152
	Transformer (test-fold epoch)	0.794 (0.766–0.819)	0.585 (0.544–0.636)	0.161
	CapsNet (test-fold epoch)	0.761 (0.735–0.788)	0.559 (0.519–0.608)	0.202
	Transformer–CapsNet (test-fold epoch)	0.793 (0.767–0.818)	0.588 (0.545–0.634)	0.177
Radiomics+clinical (concat)	Random Forest (leakage-safe, reference)	0.828 (0.805–0.851)	0.634 (0.591–0.683)	0.147
	Transformer (test-fold epoch)	0.837 (0.815–0.860)	0.649 (0.606–0.697)	0.154
	CapsNet (test-fold epoch)	0.831 (0.809–0.853)	0.646 (0.605–0.691)	0.170
	Transformer–CapsNet (test-fold epoch)	0.828 (0.806–0.852)	0.652 (0.612–0.696)	0.166
Radiomics+clinical (dual)	Random Forest (leakage-safe, reference)	0.828 (0.805–0.851)	0.634 (0.591–0.683)	0.147
	Dual Transformer (test-fold epoch)	0.843 (0.821–0.864)	0.671 (0.628–0.716)	0.154
	Dual CapsNet (test-fold epoch)	0.840 (0.817–0.861)	0.661 (0.620–0.706)	0.151
	Dual Transformer–CapsNet (test-fold epoch)	0.829 (0.806–0.851)	0.621 (0.581–0.673)	0.171